

 **AWS Storage Essentials**

Amazon S3, EBS, and EFS

AWS Certified Machine Learning Associate – Part 3

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Understanding AWS Storage Models

How Data is Accessed

Cloud storage is not "one size fits all." Different applications interact with data in fundamentally different ways—via web API calls, direct block-level disk reads, or hierarchical file system paths.

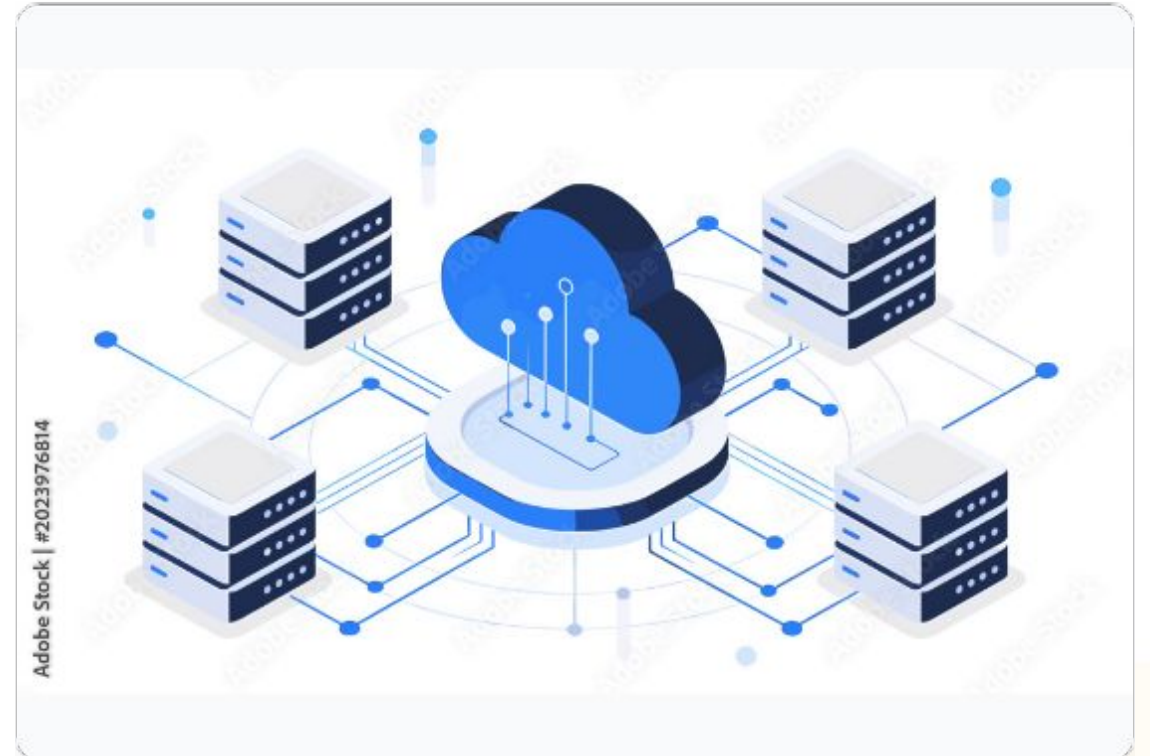
Connection Architecture

Storage choice depends heavily on who needs access: a single virtual server (EC2), thousands of distributed compute instances concurrently, or millions of internet users across the globe.

Amazon S3: Object Storage

Internet-Scale Object Store

- ✓ **Data Structure:** Key-value store (buckets and objects with customizable metadata).
- ✓ **Access Protocol:** Universal HTTP/HTTPS REST APIs (accessible anywhere).
- ✓ **Scale & Durability:** Unlimited scale with 11 9s (99.999999999%) of durability.
- ✓ **Best Use Cases:** Media files, static websites, data lakes, backups, and archives.



Amazon EBS: Block Storage

Virtual Hard Drives for EC2

Low-Latency Block Volumes: Formatted like a raw hard disk drive attached to a single EC2 instance within an Availability Zone.

Sub-Millisecond Performance: Optimized for high-throughput, IOPS-heavy workloads like relational databases (MySQL, PostgreSQL) and boot drives.

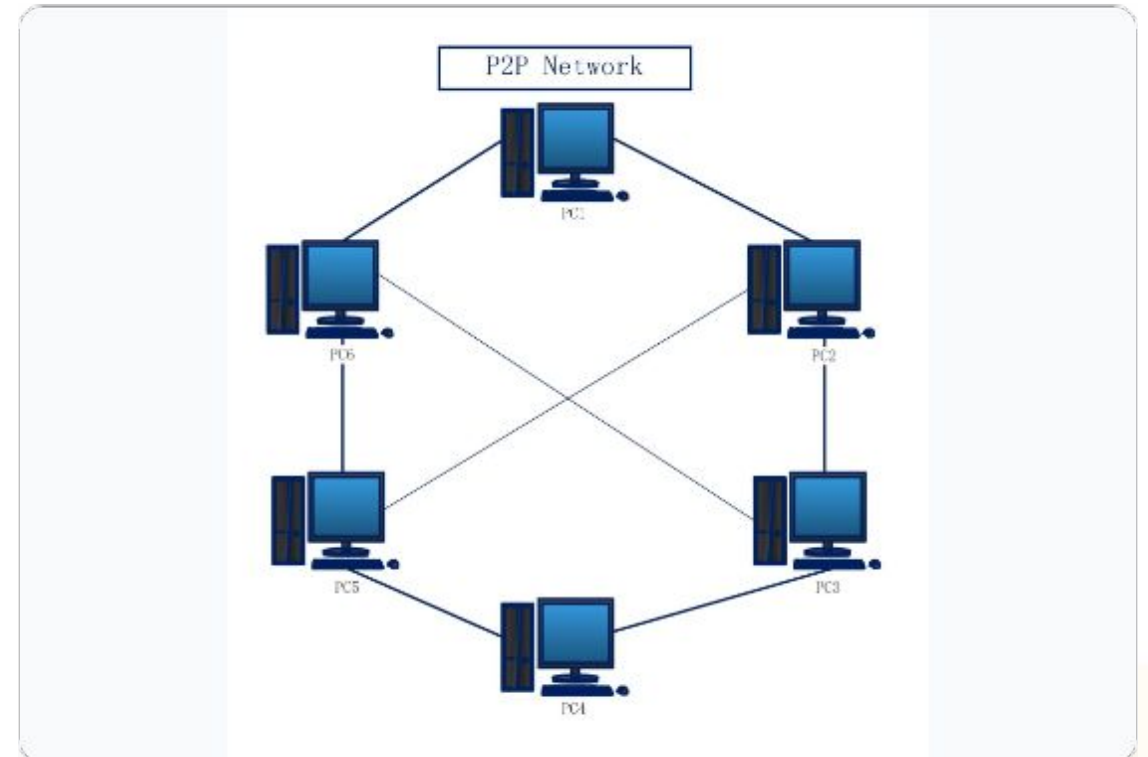
Persistence & Snapshots: Persists data independently of instance lifecycle and enables point-in-time backups to S3.



Amazon EFS: Shared File Storage

Elastic Network File System (NFS)

- ✓ **Shared Access:** Mountable concurrently by hundreds of EC2 instances and containers.
- ✓ **POSIX Standard:** Traditional file directory hierarchy with standard permission locks.
- ✓ **Serverless & Elastic:** Automatically grows and shrinks as files are added or removed without provisioning.
- ✓ **Best Use Cases:** Content management (WordPress), container storage, CI/CD tools, shared home directories.



Storage Service Comparison

Feature	Amazon S3	Amazon EBS	Amazon EFS
Storage Type	Object	Block	File (NFS)
Access Pattern	HTTP/HTTPS REST API	Block Storage (EC2 disk)	POSIX Network Mount
Concurrency	Global / Millions of Users	Single Instance (AZ specific)	1,000s of EC2 Instances
Performance	High Throughput / Variable Latency	Lowest Latency (Sub-ms)	Scalable Throughput & Latency
Cost Structure	Lowest Cost per GB	Provisioned Capacity	Pay for Actual Usage

How to Choose the Right Service



Choose EBS

When you need a fast, low-latency boot drive or persistent block storage for a database on a single EC2 instance.



Choose EFS

When multiple servers or containers need concurrent, shared access to a standard, elastic file system like WordPress uploads.



Choose S3

When storing unstructured data (media, logs, backups) accessible directly over the web or via API calls with infinite scale.

Key Takeaways

Matching your workload access pattern to the right AWS storage type optimizes both performance and cloud spend.